

地球化学专业培养方案

专业名称与代码: 地球化学 070902

专业培养目标: 本专业培养适应国家需求和学科发展需要的德、智、体、美、劳全面发展的地球化学及相关学科高级专门人才。毕业生具有扎实的地球化学专业理论和较强的实践动手能力,具有坚实的数理化基础及较强的创新能力和科学素养,能适应21世纪地球科学发展的需求和国家在资源、能源、环境、灾害、国土规划以及国民经济等相关领域对地球化学人才的需求。

专业毕业要求:

1. 数理基础与分析能力: 具有较扎实的数、理、化基础和较强的逻辑思维能力;具备英语听、说、读、写及对外交流的基本能力;具有一定的社会科学知识和管理能力;掌握一定的计算机基础知识和数据处理能力,并能应用于地球化学研究。

2. 人文与美学素养: 具有坚定的政治立场和强烈的社会责任感;具有良好的心理素质、道德修养、团队精神和一定的组织管理能力;修学人文社会科学和艺术类课程,以适应社会发展对地球化学人才多方面的需要,既可在本专业或交叉专业从事基础理论研究,也能胜任应用和管理等领域的工作,并具有一定的艺术修养。

3. 地球系统科学知识: 掌握地球科学理论基础知识,建立地球系统科学理念,具备地质和地球化学思维;具有在固体地球化学、勘查地球化学和环境地球化学等学科方向从事工作的基本专业知识。

4. 实践与交流能力: 掌握解决地球化学领域科学问题的基本方法和常用技术手段;掌握物质成分分析测试技术及基本的地球科学实验技术和野外工作技能;能够应用地球化学的思维和方法解决固体地球化学、勘查地球化学和环境地球化学等学科领域的科学问题,初步具备在上述领域从事科学研究的能力。

5. 国际视野与科研能力: 了解学科历史和国际发展前沿;具有从事地球化学研究和在环境保护、资源开发、能源利用及找矿勘查等方面开展工作的潜在能力,以满足国民经济和社会发展的需求,实现地球化学专业的人才储备;了解矿产资源勘探和环境保护等方面的政策法规;具有艰苦朴素、求真务实的专业精神;能够根据自身发展和社会需求积极开展自主学习,具备终身学习意识;在了解学科前沿及国家目标的基础上,积极开展创新性实践与探索。

毕业要求实现及途径：

序号	毕业要求	实现途径（教学过程）
1	数理基础和分析能力： 具有较坚实的数、理、化、外语、计算机的基础和分析能力。	① 课堂教学： 高等数学 B、线性代数 B、概率论与数理统计 B、大学物理、大学化学，大学英语，Python 语言程序设计等。 ② 课外学习： Python 语言课程设计等。
2	人文与美学素养： 具有坚定的政治立场和强烈的社会责任感，能胜任应用和管理等领域的工作，并具有一定的艺术修养。	① 课堂教学： 思想道德与法治、中国近现代史纲要、马克思主义基本原理、毛泽东思想和中国特色社会主义理论体系概论、习近平新时代中国特色社会主义思想概论等。 ② 课外学习： 人文讲座、社会实践等。
3	地球系统科学知识： 掌握地球科学理论基础知识，建立地球系统科学理念，具备地质和地球化学思维；具有在固体地球化学、勘查地球化学和环境地球化学等学科方向从事工作的基本专业知识。	① 课堂教学： 地球系统科学导论、地球化学、勘查地球化学、物理化学基础、地球物理学概论、矿物学导论、岩石学导论、地球生物学、沉积地球系统科学、构造地质学、地貌学与第四纪地质学等。 ② 课外学习： 地质认识实习（北戴河）、地质技能实习（周口店）、综合地球化学实习（秭归）、课余科研立项、文献阅读等。
4	实践与交流能力： 掌握解决地球化学领域科学问题的基本方法和常用技术手段；掌握物质成分分析测试技术及基本的地球科学实验技术和野外工作技能；能够应用地球化学的思维和方法解决固体地球化学、勘查地球化学和环境地球化学等学科领域的科学问题，初步具备在上述领域从事科学研究的能力。	① 课堂教学： 微量元素地球化学、同位素地球化学、勘查地球化学、环境地球化学、计算地球化学和大数据、分析地球化学、有机地球化学、土壤与土壤化学、地球化学数字模拟和制图、地球化学与人体健康、高阶地球化学主题讨论、地球化学前沿方法与技术主题研讨、地学科技论文写作等。 ② 课外学习： 地质认识实习（北戴河）、地质技能实习（周口店）、综合地球化学实习（秭归）、课余科研立项，文献阅读，学术报告与交流，毕业实习及论文设计等。
5	国际视野与科研能力： 了解学科历史和国际发展前沿；具有从事地球化学研究和在环境保护、资源开发、能源利用及找矿勘查等方面开展工作的潜在能力，以满足国民经济和社会发展的需求，实现地球化学专业的人才储备；了解矿产资源勘探和环境保护等方面的政策法规；具有艰苦朴素、求真务实的专业精神；能够根据自身发展和社会需求积极开展自主学习，具备终身学习意识；在了解学科前沿及国家目标的基础上，积极开展创新性实践与探索。	① 课堂教学： 基础课和专业课实践以及研讨部分，实践教学部分的自选课题部分，环境法规等。 ② 课外学习： 社会调查、科研立项、创新创业课程、学科竞赛等。

主干学科: 地球化学、勘查地球化学、地质学

专业核心课程: 地球化学、勘查地球化学、物理化学基础、地球系统科学导论、矿物学导论、岩石学导论、地球生物学、沉积地球系统科学、构造地质学、地貌学与第四纪地质学、地球物理学概论

主要专业实验: 地球化学样品分析实验、地球化学数据处理与应用实验、矿物鉴定、岩石薄片鉴定、古生物鉴定、地史学实习、构造分析

主要实践性教学环节: 地质认识实习（北戴河）、地质技能实习（周口店）、综合地球化学实习（秭归），专业教学实习、计算机程序设计、毕业生产实习、毕业论文

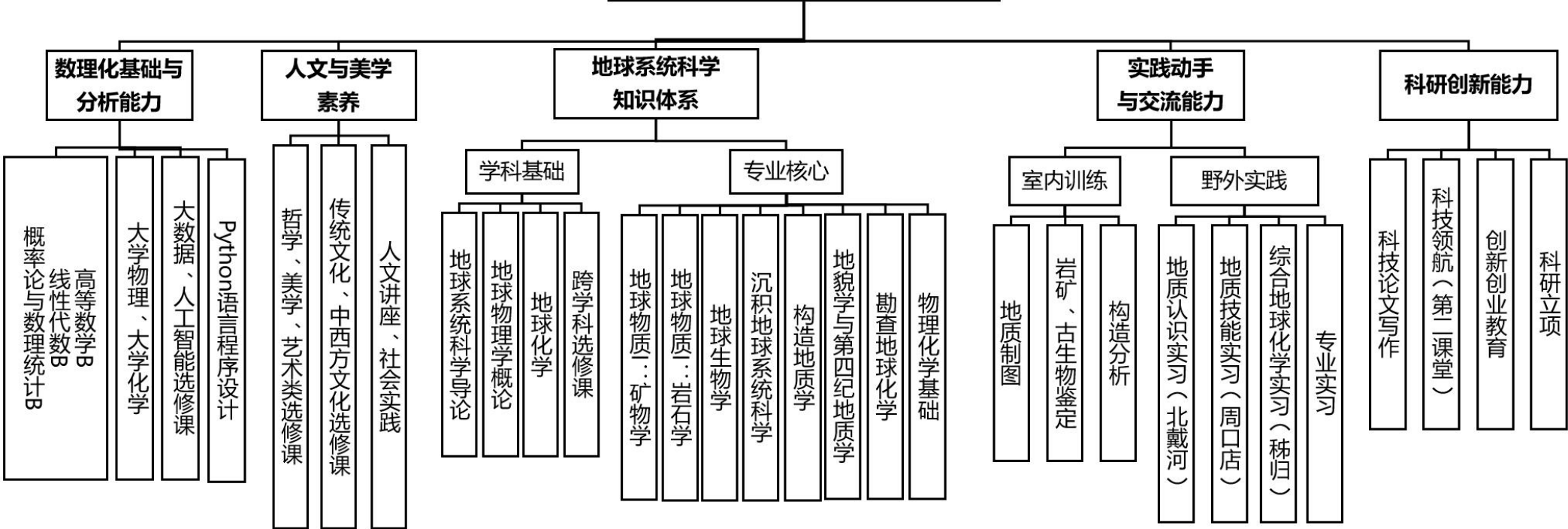
毕业学分要求: 167 学分

学制与学位: 四年，理学学士

本专业学生可以辅修的其他专业: 环境科学、地质工程、海洋科学

相近专业: 地质学

地球化学专业培养目标及定位



Program for Geochemistry Major

Specialty and Code: Geochemistry 070902

Education Objectives: The overall goal is to train high-level geochemical talents with well-balanced moral, intellectual and physical qualities that meet the demands of the socialist marketing economy. Graduates should master the fundamental theories of geology and practical ability, a solid mathematical, physical, and geochemical basis, good practical skills in computer and foreign language, and a strong sense of innovation and scientific literacy. Some graduates can continue to pursue postgraduate studies and become specialized personnel in research institutions and universities engaged in earth science research. Undergraduates of geology can adapt to the development of earth sciences in the 21st century and the country's needs for geochemical professionals in resources, environment, geological hazards, territorial planning and other related fields of the national economy.

Graduation Requirements:

1. Mathematical and physical fundamentals and analytical ability: Students should have a solid foundation in mathematics, physics, and chemistry, as well as strong logical thinking skills. Students should have basic English skills in listening, speaking, reading, writing, and communicating with others. Students should have a certain level of knowledge in social sciences and management. Students should have a certain level of computer literacy and data processing skills and can apply them to geochemical research.

2. Humanities and aesthetic literacy: Students should have a firm political stance and a strong sense of social responsibility. Students should have good psychological qualities, moral cultivation, teamwork spirit, and certain organizational and management abilities. Students should study courses in humanities, social sciences, and art to meet the diverse needs of society for geochemical talent. This enables them to engage in basic theoretical research in their own major or interdisciplinary fields, as well as to work in areas such as management and application, with a certain level of artistic accomplishment.

3. Earth system science knowledge: Mastering the theoretical foundation of earth science and establishing the concept of earth system science, and possessing geological and geochemical thinking; Students should have the basic professional

knowledge to work in disciplines such as solid earth geochemistry, exploration geochemistry, and environmental geochemistry.

4. Practical and communication skills: Master the basic methods and common technical means to solve scientific problems in the field of geochemistry; master material composition analysis and testing techniques, basic experimental techniques in earth science, and fieldwork skills. Students can apply geochemical thinking and methods to solve scientific problems in disciplines such as solid earth geochemistry, exploration geochemistry, and environmental geochemistry, and have the preliminary ability to engage in scientific research in the above-mentioned fields.

5. International perspective and research ability: Students should understand the history and international frontiers of the discipline. Students should have the potential to engage in geochemical research and work in areas such as environmental protection, resource development, energy utilization, and mineral exploration to meet the needs of national economic and social development and to realize the talent reserve of geochemistry. Students can understand policies and regulations related to mineral resource exploration and environmental protection. Students should possess a professional spirit of hard work, simplicity, seeking truth, and pragmatism. Students can actively engage in self-learning based on personal development and social needs and have a lifelong learning awareness. Based on understanding the forefront of the discipline and national goals, students can actively engage in innovative practices and exploration.

Graduation Requirements and Pathways:

No.	Graduation Requirements	Pathways (Teaching Process)
1	Mathematical and physical fundamentals and analytical ability: Students should have a solid foundation of mathematics, physics, and chemistry, as well as English and computer skills.	①Classroom Teaching: Advanced Mathematics B, Linear Algebra B, Probability Theory and Mathematical Statistics B, College Physics, College Chemistry, College English, Python Language Programming, etc. ②Extracurricular Learning: Computer Language C Course Design
2	Humanities and aesthetic literacy: Students should have a firm political stance and a strong sense of social responsibility, with a certain level of artistic accomplishment. Students can engage in areas such as management and application.	①Classroom Teaching: Ideological morality and rule of law, The Essentials of Modern Chinese History, Principles of Marxism, Introduction to Mao Tse-tung thought and the theoretical system of Socialism with Chinese Characteristics, Introduction to Xi Jinping thought on socialism with Chinese characteristics for a new era, etc. ②Extracurricular Learning: Humanities lecture, social practice, etc.

No.	Graduation Requirements	Pathways (Teaching Process)
3	Earth system science knowledge: Mastering the theoretical foundation in earth science and possessing geological and geochemical thinking; Students should have the basic professional knowledge to work in disciplines such as solid earth geochemistry, exploration geochemistry, and environmental geochemistry	①Classroom Teaching: Introduction to Earth system science, Geochemistry, Exploration Geochemistry, Introduction of Geophysics, Earth Materials: Mineralogy & Petrology, Geobiology, Sedimentary Earth system Science, Structural Geology and Tectonics, Geomorphology and Quaternary Geology, etc. ②Extracurricular Learning: Primary Field Training in Beidaihe, Geological Field training in Zhoukoudian, Geochemical Field Training in Zigui, After-school research projects, Literature reading, etc.
4	Practical and communication skills: Master the basic methods and common technical means to solve scientific problems in the field of geochemistry; master material composition analysis and testing techniques, basic experimental techniques in earth science, and fieldwork skills. Students can apply geochemical thinking and methods to solve scientific problems in disciplines such as solid earth geochemistry, exploration geochemistry, and environmental geochemistry, and have the preliminary ability to engage in scientific research in the above-mentioned fields.	①Classroom Teaching: Trace Element Geochemistry, Isotope Geochemistry, Exploration Geochemistry, Fundamentals of Physical Chemistry, Environmental Geochemistry, Computational Geochemistry and Big Data, Analytical Geochemistry, Organic Geochemistry, Pedology and Padochemistry, Geochemical Digital Simulation and Mapping, Geochemistry and Human Health, Advanced Geochemistry Seminar, Seminar on Geochemical Methods and Techniques, Geoscientific Writing, etc. ②Extracurricular Learning: Primary Field Training in Beidaihe, Geological Field training in Zhoukoudian, Geochemical Field Training in Zigui, After-school research projects, Literature reading, Academic reports and communications, Practice for graduation, Undergraduate thesis design, etc.
5	International perspective and research ability: Students should understand the history and international frontiers of the discipline. Students should have the potential to engage in geochemical research and work in areas such as environmental protection, resource development, energy utilization, and mineral exploration to meet the needs of national economic and social development and to realize the talent reserve of geochemistry. Students can understand policies and regulations related to mineral resource exploration and environmental protection. Students should possess a professional spirit of hard work, simplicity, seeking truth, and pragmatism. Student can actively engage in self-learning based on personal development and social needs and have a lifelong learning awareness. Based on understanding the forefront of the discipline and national goals, students can actively engage in innovative practices and exploration.	①Classroom Teaching: Practice and discussion parts of basic course and specialized course; Self-selected topics of practical training, Environmental Law, Geological Hazards Control, etc. ②Extracurricular Learning: social investigations, research projects, innovation and entrepreneurship courses, academic competitions, etc.

Major Disciplines: Geochemistry; Exploration Geochemistry; Geology

Core Courses: Geochemistry, Exploration Geochemistry, Fundamentals of Physical Chemistry, Introduction to Earth System Science, Introduction to Petrology, Introduction to Petrology, Geobiology, Sedimentary Earth System Science, Structural Geology and Tectonics, Geomorphology and Quaternary Geology, Introduction of Geophysics

Lab Experiments: Analytical experiments of geochemical samples; Computer operation experiments related to geochemical data processing; Mineral Identification; Fossil Identification; Tectonic Analysis; Advanced Analytical Techniques.

Practical Work: Primary Field Training in Beidaihe (Hebei Province, North China); Geological Mapping Field Training in Zhoukoudian (Beijing, North China); Comprehensive Geochemical Field Training in Zigui (Hubei Province, South China); Professional Field Training; Computer Programming; Practice for Dissertation.

Required Credits for Graduation: 167

Duration& Degree Granted: Four years, Bachelor of Science

Recommended Minors: Environmental Science, Geological Engineering, Marine Science

Related Specialties: Geology

地球化学专业课程教学计划表

Course Descriptions of Geochemistry Major

课程类别 Category		课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours					先修课程 Prerequisite courses	学期学分分配 Semester Credits								
						课内学时		课外学时				一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th	
						讲课 Lec.	课内实验 Lab	实验/科研实践 Lab/Re s.	研讨 Dis	素质拓展 Exp										
通识教育课 Liberal Education Courses	必修 Compulsory	12007800	马克思主义基本原理 Principles of Marxism	3	48	48						3								
		12008100	毛泽东思想和中国特色社会主义理论体系概论 Introduction to Mao Tse-tung Thought & the Theoretical System of Socialism with Chinese Characteristics	2	32	32									2					
		12008000	习近平新时代中国特色社会主义思想概论 Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	3	48	48									3					
		1171800	中国近现代史纲要 Essentials of Modern Chinese History	2	32	32								2						
		12007900	思想道德与法治 Ideological Morality & Rule of Law	3	48	32	16						3							
		12005300	形势与政策 Situation & Policy	2	32	32							每学期平均分配							
		113076*0	体育 Physical Education	4	144	144							1	1	1	1				
		109234*0	大学英语 College English	9	144	144					48		3	3	3					
		14300300	军事理论 Military Theory	2	36	36							2							
		12008200	劳动教育 Labor Education	1	16	16							1							
		21624800	大学语文 B College Chinese B	2	32	32								2						
		20123000	逻辑学导论 Logic	2	32	32							2							
		21801000	哲学基础 Foundation of Philosophy	2	32	32							2							
	选修 Elective		经典阅读与分享（1-7 学期）（必选） Literature Reading & Sharing	1					16									1		
			人文讲座（1-7 学期）（必选） Humanities Seminars	1					16									1		

地球化学专业

课程类别 Category		课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours					先修课程 Prerequisite courses	学期学分分配 Semester Credits								
						课内学时		课外学时				一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th	
						讲课 Lec.	课内实验 Lab	实验/科研实践 Lab/Res.	研讨 Dis	素质拓展 Exp										
		总计 12 学分。包括经典阅读与分享、人文讲座必修课程（各 1 学分），美育、心理健康教育课程（各不少于 2 学分），跨学科选修课不低于 4 学分。		12	192	192			32			4					4	4		
		小计 Sum		49	868	852	16		32	48		21	8	4	6		4	5	1	
		2194 7400	Python 语言程序设计 Python Language Programming	2.5	40	40		16						2.5						
		2121 27*2	高等数学 B Advanced Mathematics B	10	160	160						4	6							
		2121 2802	线性代数 B Linear Algebra B	2.5	40	40								2.5						
		2121 3502	概率论与数理统计 B Probability Theory & Mathematical Statistics B	2.5	40	40									2.5					
		2121 30*1	大学物理 A College Physics A	8	128	128							4	4						
		2121 69*1	物理实验 A Physical Experiments A	2	64	4	60						1	1						
		2032 69*1	大学化学 A College Chemistry A	5.5	88	88							2.5	3						
		2032 70*1	大学化学实验 A College Chemistry Experiment A	2	48		48						1	1						
			小计 Sum	35	608	500	108	16					4	14.5	11.5	2.5	2.5	0	0	0
		学科基础课 Disciplinary Fundamental Courses		2012 3500	地球系统科学导论 Introduction to Earth System Science	3	48	36	12			16		3						
2012 0200	地球化学 Geochemistry			3	48	48			16		地球物质 II、构造地质学、沉积地球系统科学				3					
2062 8000	地球物理学概论 Introduction of Geophysics			3	48	44	4				高数、大学物理、线性代数					3				
2040 6900	普通生物学 General Biology			2	32	32						2								
	小计 Sum			11	176	164	12		16	16		5	0	0	3	3	0	0	0	
2012 3710	地球物质 I：矿物学 Earth Materials I: Mineralogy			4	64	28	36	4			大学化学		4							
2012 3720	地球物质 II：岩石学 Earth Materials II: Petrology			4	64	30	34	4	4		地球物质 I			4						

中国地质大学（武汉）本科培养方案

课程类别 Category	课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours					先修课程 Prerequisite courses	学期学分分配 Semester Credits								
					课内学时		课外学时				一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th	
					讲课 Lec.	课内实验 Lab	实验/科研实践 Lab/Re s.	研讨 Dis	素质拓展 Exp										
	2012 3800	地球生物学 Geobiology	3	48	32	16		8	8	地球系统科学导论			3						
	2012 3900	沉积地球系统科学 Sedimentary Earth system Science	4	64	48	16		8	8	地球生物学、岩石学导论、构造地质学				4					
	2010 4001	构造地质学 A Structural Geology & Tectonics A	4	64	40	24				地球系统科学导论、岩石学导论				4					
	2012 4500	地貌学与第四纪地质学 Geomorphology & Quaternary Geology	2.5	40	30	10			8							2.5			
	2022 2502	矿床学 B Mineral Deposits B	2	32	32		16			岩石学导论					2				
	2011 2300	勘查地球化学 A Exploration Geochemistry A	3	48	34	14				地球化学						3			
	2012 6500	物理化学基础 Fundamentals of Physical Chemistry	3	48	48												3		
	小计 Sum		29.5	472	324	148	24	20	24			4	7	8	2	5.5	3	0	
专业选修课 Specialty Elective Courses	在学术导师指导下选择不少于 15 个学分的专业选修课，其中跨专业选修课 6 学分，专业内选修课 9 学分																		
	跨专业 Interdisciplinary	2012 4400	科技论文写作（必选） Science & Technology Paper Writing	1	16	16									1				
		2230 3700	人工智能 Artificial Intelligence	2	32	28	4									2			
		2113 0402	测量学 B Surveying B	1.5	24	24		12				1.5							
		2114 8500	地图学 Cartography	2	32	32		8								2			
		2011 1601	自然地理学 A Physical Geography A	3	48	48									3				
		2011 7700	遥感地质学 Remote Sense Geology	3	48	32	16									3			
		2020 2800	海洋学基础 Basic Oceanography	3	48	48	0												3
		2012 1600	大气科学概论 Introduction to Atmospheric Science	1.5	24	24	0											1.5	
		2110 0700	GIS 原理与应用 Principle & application of GIS	2.5	40	24	16											2.5	

地球化学专业

课程类别 Category		课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours					先修课程 Prerequisite courses	学期学分分配 Semester Credits							
						课内学时		课外学时				一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
						讲课 Lec.	课内实验 Lab	实验/科研实践 Lab/Res.	研讨 Dis	素质拓展 Exp									
专业内 Disciplinary		2050 8400	工程地质学基础 B Engineering Geology B	2.5	40	32	8											2.5	
		2022 2303	石油及天然气地质学 C Petroleum Geology C	2.5	40	40		8								2.5			
		2023 0802	矿产勘查理论与方法 B Theories Method of Mineral Resource Exploration B	2.5	40	40		8									2.5		
		2040 5303	环境评价 C Environmental Assessment C	2	32	32												2	
		2170 4500	环境法规 Environmental Law	1.5	24	24												1.5	
		2050 6000	地质灾害防治 Geological Hazards Control	2	32	32					工程地质基础						2		
		2012 6600	微量元素地球化学 Trace Element Geochemistry	1.5	24	18	6		8						1.5				
		2011 7500	同位素地球化学 Isotope Geochemistry	1.5	24	24									1.5				
		2011 7300	分析地球化学 Analytical Geochemistry	2	32	20	12								2				
		2012 6000	计算地球化学和大数据 Computational Geochemistry & Big Data	2	32	24	8									2			
		2010 9100	环境地球化学 Environmental Geochemistry	2	32	24	8									2			
		2011 0400	有机地球化学 Organic Geochemistry	2	32	26	6									2			
		2010 6200	土壤与土壤化学 Pedology & Pedochemistry	2	32	24	8								2				
		2012 6100	地球化学数值模拟和制图 Geochemical Digital Simulation & Mapping	2	32	8	24										2		
		2012 6200	地球化学与人体健康 Geochemistry & Human Health	1.5	24	24											1.5		
		2012 6300	高阶地球化学主题研讨 Advanced Geochemistry Seminar	1.5	24	12	12										1.5		
		2012 6400	地球化学前沿方法与技术主题研讨 Seminar on Geochemical Methods & Techniques	1.5	24	12	12										1.5		
		2011 6300	区域地质调查技术与方法 Methods & Techniques in Regional Geological Survey	2	32	16	16									2			
		2011 3800	全球变化 A Global Change A	3	48	40	8									3			
		2011 9200	行星地质学导论 Introduction to Planetary Geology	3	48	40	8								3				

中国地质大学（武汉）本科培养方案

课程类别 Category		课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours					先修课程 Prerequisite courses	学期学分分配 Semester Credits							
						课内学时		课外学时				一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
						讲课 Lec.	课内实验 Lab	实验/科研实践 Lab/Re s.	研讨 Dis	素质拓展 Exp									
		20124800	应用矿物学 Applied Mineralogy	3	48	32	16										3		
		20118900	地震地质学 Earthquake Geology	3	48	48										3			
		小计 Sum		15	240	240						0	0	0	0	4	4	4	3
合计 Sub-total				139.5	2364	2080	284	40	68	88		30	26.5	22.5	19.5	11.5	13.5	12	4
实践环节 Practical Work	44300400	军事训练 Military Training	2	2周							2								
	42008300	劳动实践 Labor practice	1	4周													1		
	41947500	Python 语言课程设计 Course Design for Python Language	1.5	1.5周									1.5						
	40115200	地质认识实习（北戴河） Primary Field Training in Beidaihe	2	2周								2							
	40124000	地质技能实习（周口店） Geological Field training in Zhoukoudian	4	4周										4					
	40117900	综合地球化学实习（秭归） Geochemical Field Training in Zigui	3	3周													3		
	40118500	毕业生产实习 Practice for Graduation	4	4周														4	
	40125500	毕业论文（设计） Thesis for Graduation	4	4周															4
		小计 Sum		21.5	24.5周							2	2	1.5	4	0	4	4	4
创新创业自主学习 Freedom study	ZZ35000S	社会调查 Social Investigation	2	2周												2			
		科研立项 Scientific Research	2	4周													2		
		其他(含创新创业课程、学科竞赛、发明创造、科研报告) Others (Start-up, Contest, Invention, Innovation & Research Presentation)	2	2周															2
		小计 Sum		6	8周							0	0	0	0	2	2	0	2
总计 Total				167	2364+32.5周	2080	284	40	68	88		32	28.5	24	23.5	13.5	19.5	16	10

注：全英课程须在课程名称后打*标出，通识教育选修课学分未列入具体学期，学院须根据学校创新创业自主学习学分认定一览表制订实施细则。

地球化学专业课程分类统计

Course Classifications and Statistics of Geochemistry Major

课程类别 统计	通识教育课程 Liberal Education Courses		理科基础课+学科基础 课 Platform & Disciplinary Fundamental Courses	专业核心课 Main Specialty Courses	专业选修课 Specialty Elective Courses	实践环 节 Practica l Work	创新创业自 主学习 Freedom Study	学时总计 Total Hours	学分总计 Total Credits
	必修	选修							
课内总学 时 /学分	676//37	192/12	784/46	472/29.5	240/15	24.5 周 /21.5	8 周/6	2364+32. 5 周	167
学分所占 比例	29.3%		27.5%	17.7%	9%	12.9%	3.6%		100%

地球化学专业辅修课程教学计划表

Course Descriptions of Geochemistry (Minor)

课程类别 Classi- fication	课程 编号 Cod e	课程名称 Course Name	学 分 Cr s	课 内 总 学 时 Hrs	学时分类 Class Hours					先修课程 Prerequisi- te courses	学期学分分配 Semester Credits							
					课内学时		课外学时				一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
					讲 课 Lec.	课 内 实 验 Lab	实 验 /科 研 实 践 Lab/Re- s.	研 讨 Dis	素 质 拓 展 Exp									
学科基础课 Disciplinary Fundamental Courses	20119600	普通地质学 Physical Geology	2.5	40	40		8								2.5			
	20222502	矿床学 B Mineral Deposits B	2	32	32		16									2		
	20603800	固体地球物理学概论 Geophysics	2.5	40	38	2										2.5		
	小计 Sum		7	112	112	0	24	0	0	0	0	0	0	2.5	0	4.5	0	
专业主干课 Main Specialty Courses	20113100	矿物岩石学 Mineralogy & Petrology	3	48	12	36									3			
	20104001	构造地质学 B Structural Geology B	3	48	30	18										3		
	20120200	地球化学 Geochemistry	3	48	48											3		
	20109100	环境地球化学 Environmental Geochemistry	2	32	32					①地球化学							2	
	20105000	勘查地球化学 B Exploration Geochemistry B	2	32	24	8				①地球化学							2	
	20117300	分析地球化学 Analytical Geochemistry	2	32	20	12				地球化学							2	
	小计 Sum		15	240	166	74					0	0	0	0	3	6	2	4
实践环节 Practical Work	40122400	综合地球化学实习（稀归）	3	3周													2	
	小计 Sum		2	3周													2	
总计 Total			25	352+3周	278	74	24	0	0	0	0	0	0	0	5.5	6	8.5	4

地球化学辅修专业课程分类统计

Course Classifications and Statistics of Geochemistry (Minor)

课程类别 统计	学科基础课 Disciplinary Fundamental Courses	专业核心课 Main Specialty Courses	实践环节 Practical Work	学时总计 Total Hours	学分总计 Total Credits
课内总学时 /学分	112/7	240/15	3 周/3	352+3 周/	25
学分所占比例	28.0%	60.0%	12.0%		100%